

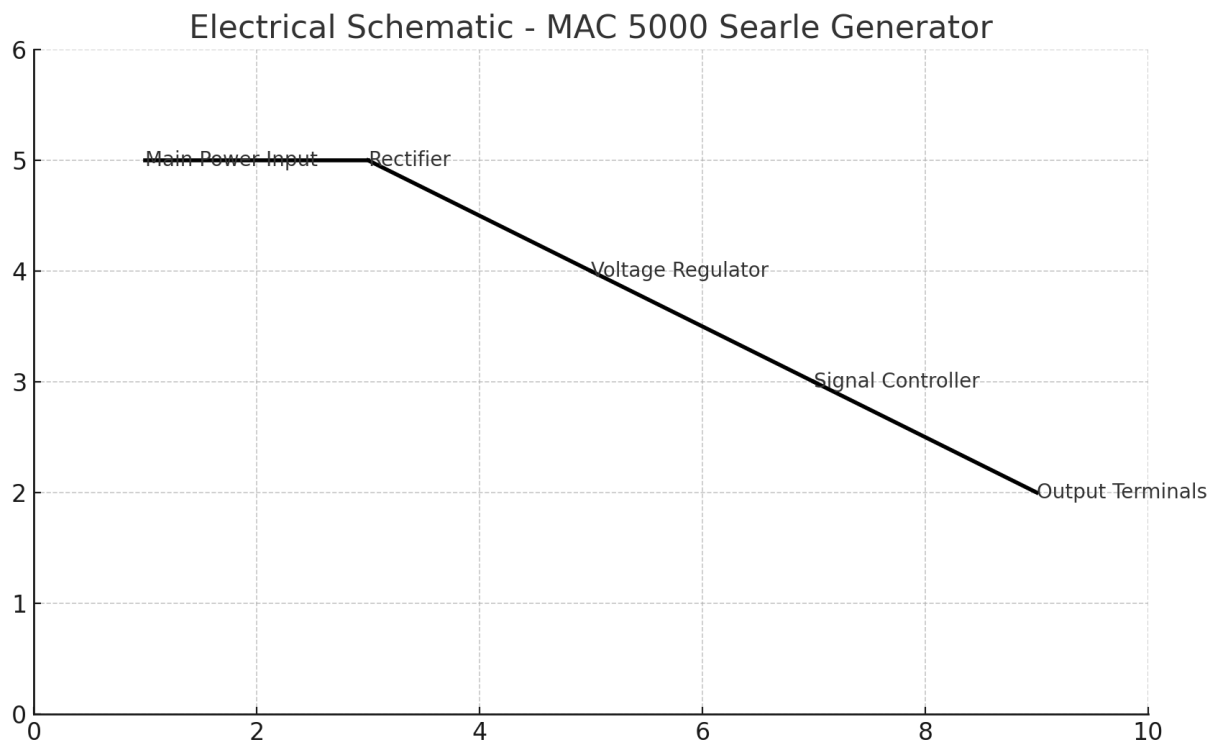
MAC 5000 Searle Generator

Final System Documentation

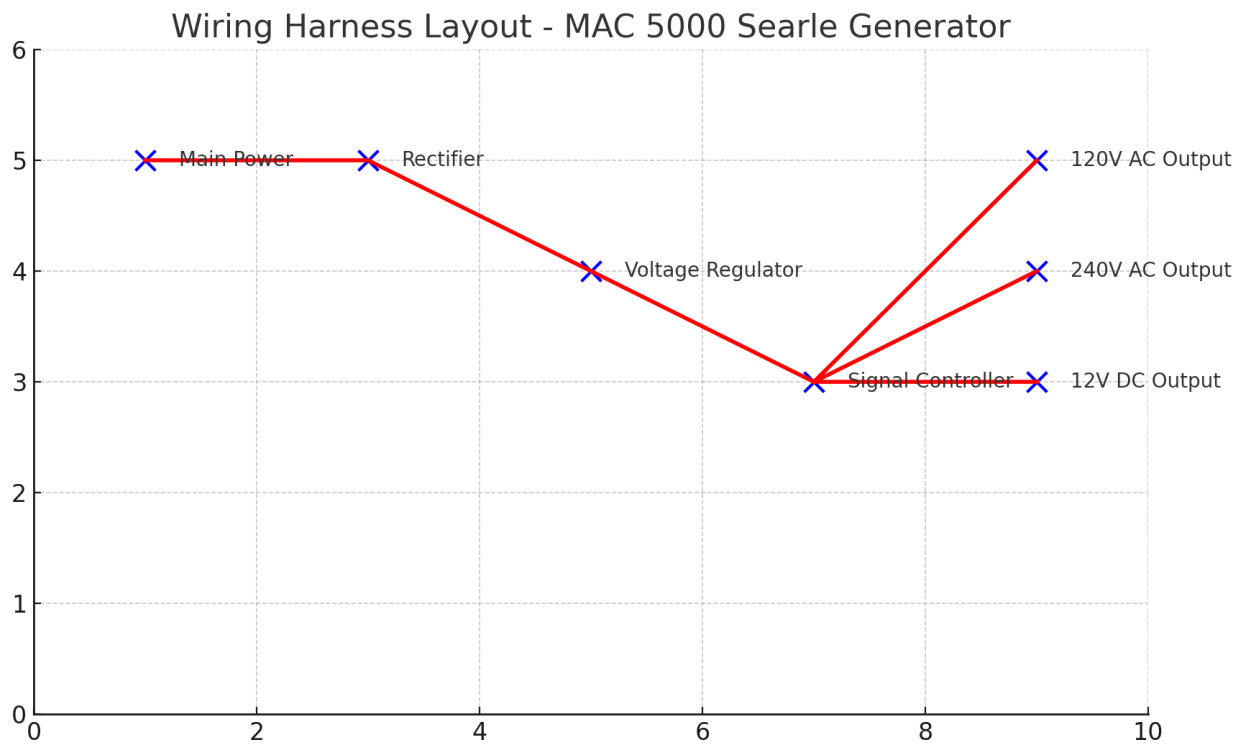
1. Introduction & System Overview

This document provides a complete guide to the MAC 5000 Searle Generator, including schematics, wiring diagrams, PCB layouts, assembly instructions, and testing procedures.

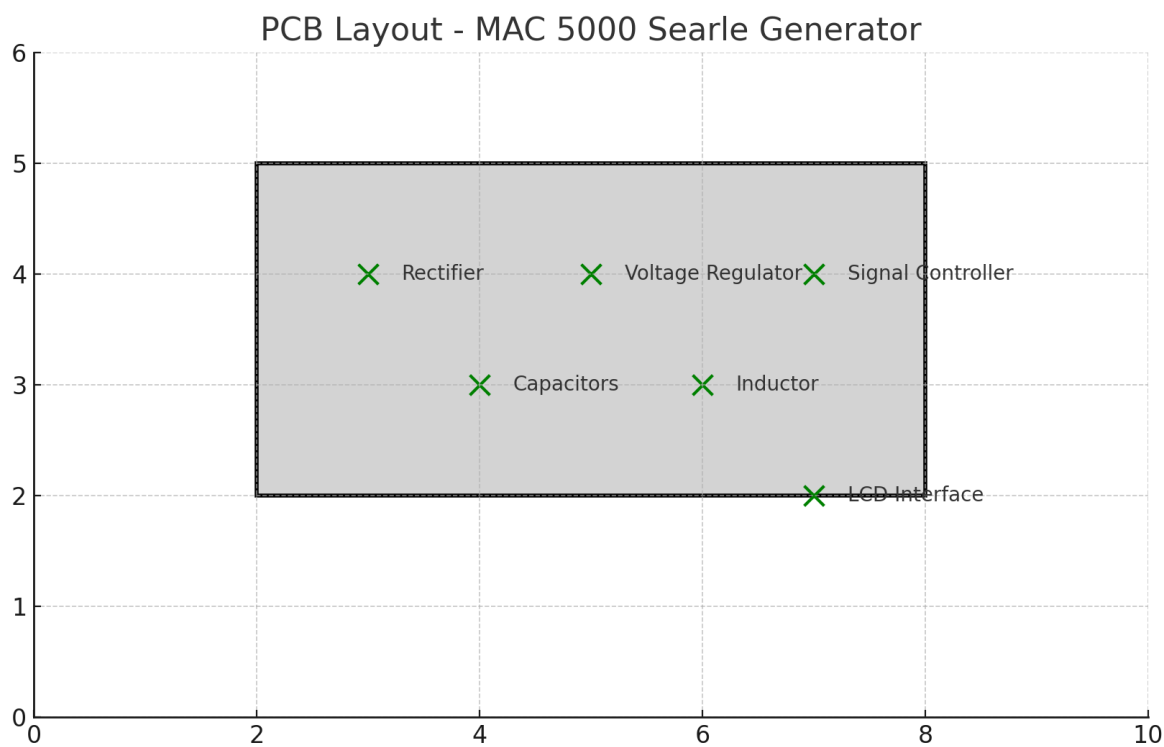
2. Full Electrical Schematic



3. Wiring Harness Layout



4. PCB Design & Layout



The PCB is designed to integrate rectifiers, voltage regulators, signal controllers, and power management components efficiently.

5. Assembly Instructions

Step-by-step instructions for assembling the MAC 5000 Searle Generator:

1. Prepare the PCB and solder components according to the provided guide.
2. Connect the wiring harness to the PCB and power distribution.
3. Secure all components within the enclosure.
4. Perform initial power-on tests before full system integration.

6. Testing & Troubleshooting Guide

Before powering the system, ensure:

- Proper soldering and secure wiring.
- No shorts between power and ground.
- Safe electrical isolation and grounding.

Use a multimeter to verify correct voltages at each output.

7. Parts List & Sourcing

The following table lists the required components, suppliers, and costs:

Rectifier Bridge (50A, 1000V)	- 1 pcs	- Mouser Electronics	- \$5.0
Voltage Regulator (LM317)	- 1 pcs	- Digi-Key	- \$3.5
Signal Controller (Microcontroller)	- 1 pcs	- SparkFun	- \$12.0
Electrolytic Capacitor (1000uF, 50V)	- 2 pcs	- Mouser Electronics	- \$1.2
Ceramic Capacitor (0.1uF, 50V)	- 4 pcs	- Digi-Key	- \$0.5
Inductor (100uH, 5A)	- 1 pcs	- Adafruit	- \$2.5
LCD Display (16x2 I2C)	- 1 pcs	- Amazon	- \$8.0
Power Switch (Toggle, 250V 10A)	- 1 pcs	- Mouser Electronics	- \$4.5
Emergency Stop Button (NC, 250V 5A)	- 1 pcs	- Digi-Key	- \$6.0
Output Terminals (Screw Type, 600V 20A)	- 3 pcs	- Amazon	- \$3.0
PCB Board (Custom Etched)	- 1 pcs	- Custom Fabrication	- \$15.0
Copper Traces (35µm Thickness)	- N/A pcs	- N/A	- \$N/A
Resistors (Assorted 1kΩ - 10kΩ)	- 5 pcs	- Mouser Electronics	- \$0.2
Diodes (1N4007)	- 4 pcs	- Digi-Key	- \$0.3
Heatsinks for Voltage Regulator & Rectifier	- 2 pcs	- Amazon	- \$3.0
Conformal Coating (Protective Layer)	- 1 pcs	- Digi-Key	- \$6.0
Wire Connectors (JST & Screw Terminals)	- 10 pcs	- Amazon	- \$0.5